USN: 1RVU23CSE394

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Exp No: 9 **Date : 12/11/2025**	**Data Warehousing in Snowflake – Transforming a Normalized Schema into a Star Schema**

Objective:

The objective of this lab is to demonstrate how a fully normalized database model can be transformed into a Star Schema in Snowflake for analytical purposes. The aim is to redesign the ClassicModels dataset, originally structured for transactional operations, into a dimensional model suitable for reporting and analytics.

Outcomes:

• Understand the structure and limitations of a normalized schema for analytical workloads.

• Design dimension and fact tables that follow the Star Schema approach.

• Identify primary business entities and convert them into dimensions.

• Identify measurable business events and convert them into fact tables.

• Understand how analytical queries benefit from a star schema.

Materials:

• Snowflake environment or SQL worksheet

• Normalized schema description and sample data

• System with internet access

• Analytical requirements for customers, products, orders, and sales patterns

Lab Procedure:

Stage 1: Problem Definition and Requirement Understanding

The ClassicModels dataset represents a traditional normalized schema containing several interconnected tables such as Customers, Orders, Employees, Offices, Products, ProductLines, and OrderDetails. These tables are designed for transactional operations (OLTP). The requirement is to restructure data into dimensions and a central fact table to support analytics.

Stage 2: Understanding the Normalized Data Model

The normalized model stores atomic, non-redundant information. However, analytical queries require multiple joins and complex navigation. This motivates conversion into a star schema for improved performance and simplified querying.

Stage 3: Designing the Star Schema

The star schema includes:

• DIM_CUSTOMER

• DIM_PRODUCT

• DIM_EMPLOYEE

• DIM_DATE

• FACT_ORDERS (central fact table with sales metrics)

This design aligns with business-friendly analytical structures.

Stage 4: Mapping Normalized Tables to Dimensions

Each normalized table maps to a dimension:

• Customers → DIM_CUSTOMER

• Products → DIM_PRODUCT

• Employees → DIM_EMPLOYEE

• Order Dates → DIM_DATE

Attributes were chosen based on analytical relevance.

Stage 5: Populating the Star Schema

Dimension tables were populated with descriptive attributes, and the fact table with order-line–level data capturing quantities, sales totals, and links to dimensions. This followed the extract–transform–load workflow.

Stage 6: Analytical Interpretation in Star Schema

The star schema made it easier to perform analysis such as total quantity ordered per customer, product-wise performance, employee contributions, and time-based sales trends. The simplified model improved query performance.

Stage 7: Verification and Validation

Validation ensured dimensions contained correct attributes, fact entries aligned with normalized data, and analytical totals matched expected outcomes.

Reflection:

This experiment demonstrated how transforming a normalized operational schema into a star schema improves analytical performance, simplifies business queries, and supports scalable reporting. The redesigned structure offers a strong foundation for dashboards and BI tools.

GitHub Link: